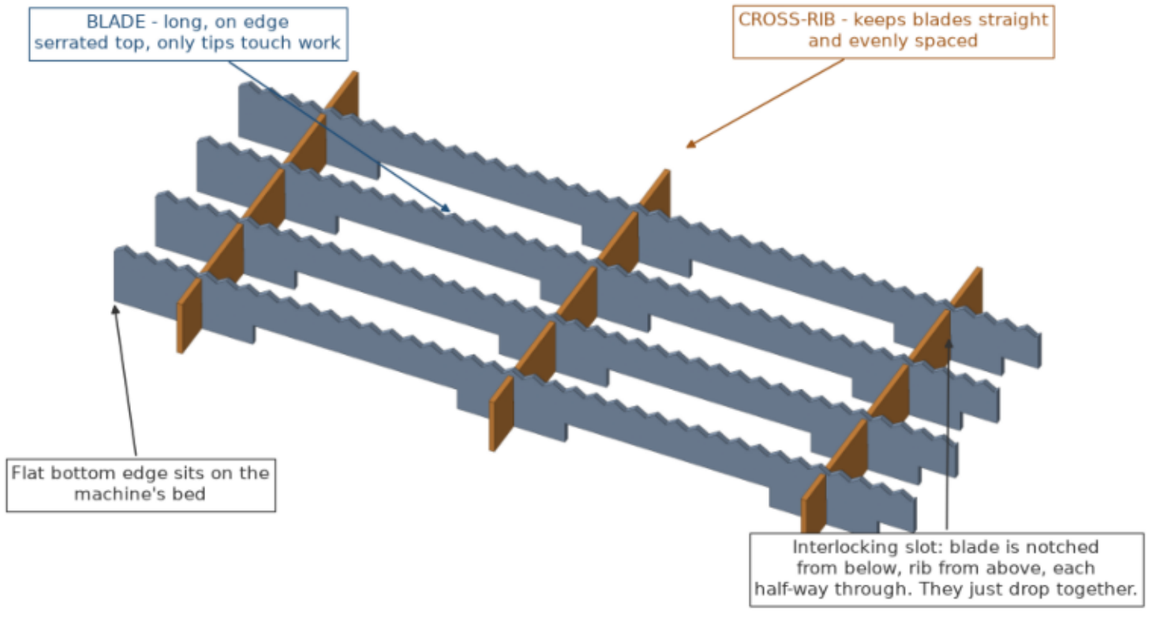
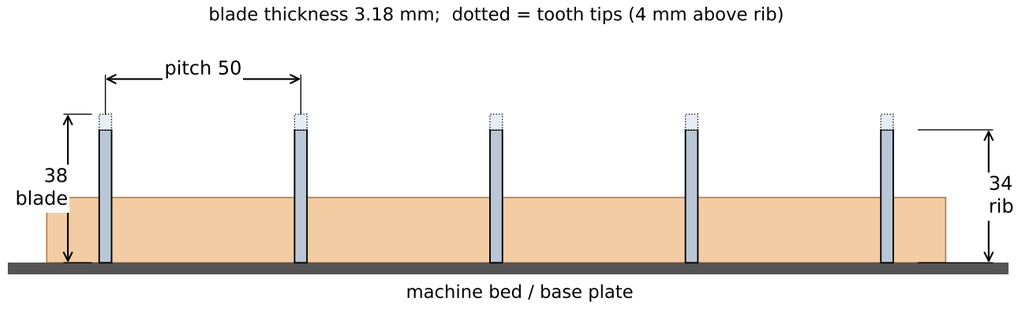
EGG-CRATE STEEL BED - ACMER X1 Ultra - assembly (page 1 of 2)

Units: mm. Material: mild steel 1/8in (3.18 mm). Laser cut, interlocking half-depth slots. Dimensions to be confirmed after measuring machine inside opening.

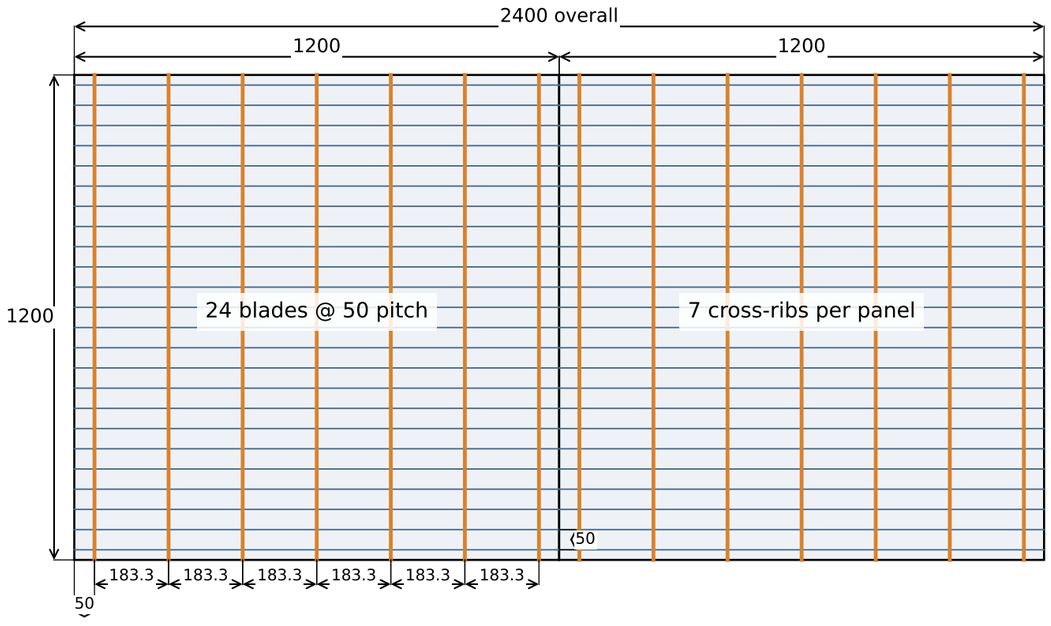


SIDE VIEW (looking along the blades) - 5 blades shown



orange = cross-rib behind the blades (it sits in the blade slots)

TOP VIEW - whole bed (2 panels, end to end)



Panel 1 / Panel 2 butt together at the seam (blades line up end to end). Ribs run across the blades.

QUANTITIES & NOTES

A BLADE qty 48 (24/panel) 1200 x 38 x 3.18 0.83 kg each

B CROSS-RIB qty 14 (7/panel) 1200 x 34 x 3.18 0.98 kg each

Total steel: 54 kg (119 lb); 27 kg per panel.

Sheet: one 1220 x 2440 mm (4x8 ft) sheet of 1/8in weighs 74 kg.

Nesting: one panel (31 parts) fits on ONE 4x8 sheet in 668 of 1220 mm width; whole bed (62 parts) needs 2 4x8 sheets (2nd sheet 4 parts) or 1 sheet of 60x120.

ASSEMBLY: ribs flat on the bench with slots up; drop blades in with slots down. No bolts or welds. Lifts out as two panels of about 27 kg each (two-person lift).

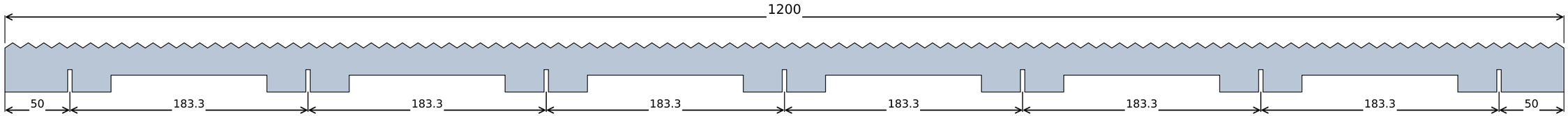
USE: diode laser (cuts drop through the grid); 4 W IR (blade tips are the only contact, flat and level); CNC uses a separate lift-out spoilboard.

TO CONFIRM: panel size 1200 x 1200 was chosen to fit a 4x8 sheet and the work area. The machine's inside opening is not published (estimated 2530 x 1470 mm) and ACMER's '44 mm' cover-to-surface figure is unexplained. Measure before ordering.

EGG-CRATE STEEL BED - part drawings (page 2 of 2)

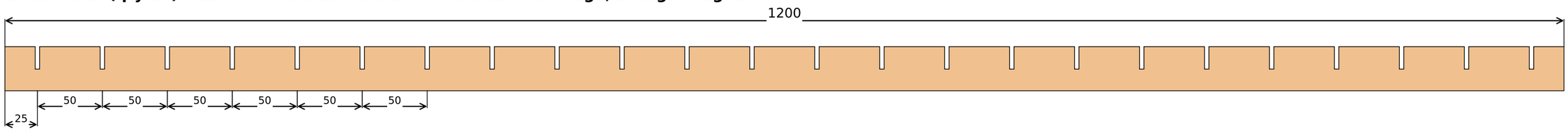
Units mm. CUT geometry = finished part outline (vendor to apply kerf compensation). Not to scale where labelled 'detail'.

PART A - BLADE (qty 48) - 1200 x 38 x 3.18 - slots cut from the FLAT BOTTOM edge, teeth on top



7 slots per blade, centres at 50.0, 233.3, 416.7, 600.0, 783.3, 966.7, 1150.0

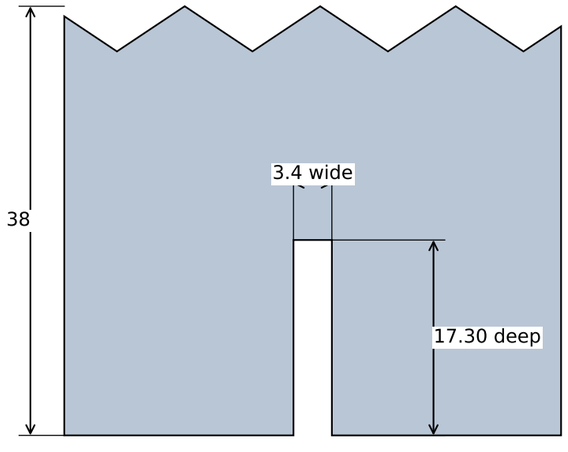
PART B - CROSS-RIB (qty 14) - 1200 x 34 x 3.18 - slots cut from the TOP edge, straight edges



24 slots per rib, 50 pitch, first centre 25 from each end

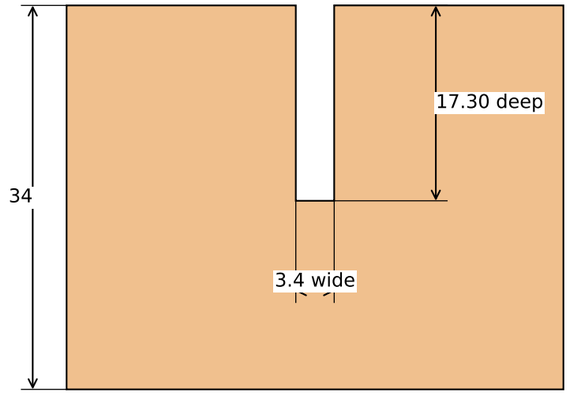
DETAIL 1 - blade slot (from bottom)

teeth: see detail 3



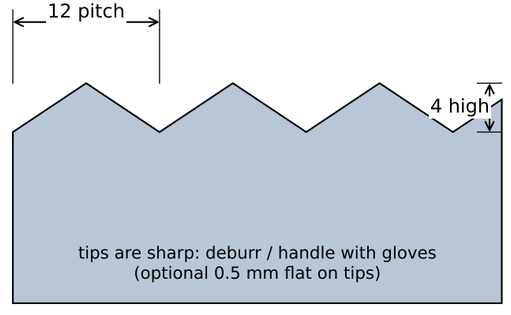
slot = 3.18 + 0.2 mm clearance, kerf to be adjusted by vendor

DETAIL 2 - rib slot (from top)



remaining web under slot 16.70

DETAIL 3 - serrated top (blades only)



NOTES

1. Material: mild steel (A36 / equivalent), 1/8in = 3.18 mm. 11 ga (3.04 mm) also OK; if 11 ga, slot width becomes 3.25 mm (regenerate with --material 11ga).
2. Slot width = thickness + 0.2 mm. Vendor to compensate kerf; widen to 3.5 if tight.
3. Tolerances: slot width +0.15/-0.05, slot position +/-0.2, length +/-0.5.
4. Deburr all edges, remove dross; leave bare steel (light oil). If coated, add coating thickness to slot width.
5. Qty: 48 x A, 14 x B. Two panels of 1200 x 1200.
6. Dimensions to be confirmed after measuring the machine inside opening.

ASSEMBLY CHECK (computed): verify: 24 blades x 7 ribs = 168 crossings per panel; blade slot centres == rib positions, rib slot centres == blade positions; vertical clearance at every crossing = 0.60 mm (>=0, no overlap)